



BUSINESS BLUEPRINT AOP AUTOMATION FOR NMD PP PLANT

Honeywell

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Business Blueprint

AOP Automation for NMD PP Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

Honeywell Automation India Ltd (“Honeywell”)



Revision History

Rev.	Date	Originated By	Description of status and changes
0.0	01-09-2024	HON	Issued for Review
1.0	28-07-2025	HON	Issued for Review (Discussed logic Updated)

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1. Introduction

The purpose of this document is to capture the detailed specifications for the Digitalized Annual Operating Plan (AOP) for NMD PP plant for NMD site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various Polyethylene plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple polypropylene (PP and PPCP) plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business as the next revised budget by the end of Jan. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for PP technology.

1.2 Reference Documents

Document ID	Revision	Title	Document
O-836243B24R01	R01	AOP Automation for O2C Proposal	 O-836243B24R01 RIL AOP Automation f
PP effective Operating Hours calculation	-	RIL AOP - PP Maintenance Screen – Reliance to Confirm on logic	

NMD PP DCT	-	Digital AOP NMD PP DCT	 NMD_PP_DCT_Rev02 .xlsx
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Table 1 Reference Documents

2. AOP Design Basis

The below table captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release post review and discussion

Table 2 AOP Functional Scope

Function-wise detailed design basis is captured in the below sections.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Consumption AOP

Steps to generate the Production and Consumption AOP are captured in the below flow diagram.

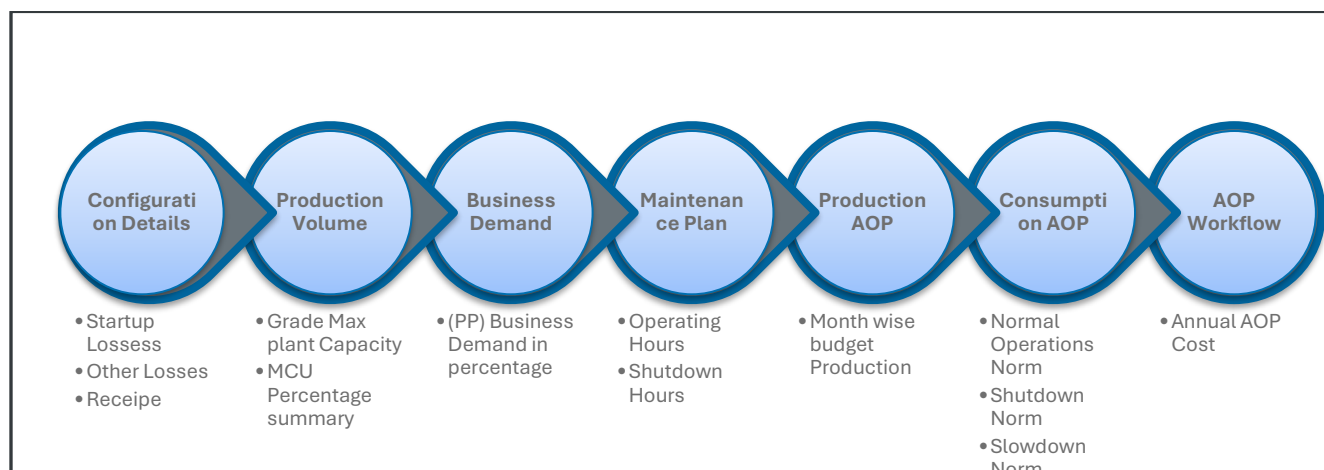


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Configuration

The configuration screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

Plant-specific data is required to calculate Production and consumption. data, Startup losses for raw material & Utilities, other losses and Recipe. There are different tabs available to configure the above parameter under the Configuration screen to support Production and Consumption Norms calculations and reports generation.

By default, the values manual entries entered for the current year will be automatically carried forward to the next year.

In configuration screen, below additional features and tabs are provided.

- **AOP historical period** basis added where Start and End date is provided to select historical period to calculate MCU and Norms data for selected period by default 05 years data will be loaded. AOP Design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- **Recipe tab** is provided to enter the catalyst chemical recipe value which will not change month wise but can vary grade wise.
- **import and export** options to import and export manual entry parameters from configuration screen are provided.
- **Startup losses** are material wise losses during the startup period.
- **Other losses** are material wise other losses.
- **Shutdown Rate (Per HR)** is rate of losses during shutdown period.

Historical Period Basis.

Production & Norms Basis

Year: Vertical: Site: Plant:

- Production & Norms Basis**
- Shutdown/TA Activities
- Slowdown Activities
- Production Target
- Business Demand
- Maintenance Details
- Month wise Production plan
- Steady State Consumption (Norm/Quantity)
- Shutdown Consumption (Quantity)
- Slowdown Consumption (Norm/Quantity)
- Overall AOP Consumption (Norm/Quantity)
- Feed Stock Availability

PP / NMD / PP

AOP Historical Period Basis

Start Date: End Date: (Last refreshed data on: 25-07-2025 02:49 PM for the period from 01-12-2023 to 30-11-2024)

AOP Design Basis

PP Manual Input configuration screens

❖ Startup Losses

Startup Losses															Other Losses	Shutdown Rate (Per Hr)	Receipte	
															Save			
Particu...	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark				
▼ Raw Material																		
PROPYLENE (P...	MT	12	11	9	8	7	8	6	6	5	9	9	10	test1				
ETHYLENE	MT	13	12	11	15	15	15	15	15	15	15	15	15	test4				
PROPYLENE- C...	MT	11	12	12	12	12	12	12	12	12	12	12	12	test2				
▼ Utility Consumption																		
COMPRESSED ...	NM3	11	14	12	10	12	14	11	12	17	11	21	13	Test4				
D M Water	M3	12	12	11	12	15	14	19	13	12	10	12	12	Test5				
Power_Dis	KWH	13	11	10	11	16	12	10	12	12	15	17	11	Test44				
LP Steam_Dis	MT	14	10	12	14	15	11	14	11	11	12	12	10	Test7				
Effluent Treated	M3	15	23	14	15	13	14	12	12	12	15	19	12	Test40				
Cooling Water 2	KM3	16	25	15	15	12	12	15	14	15	23	19	17	Test14				
Water	M3	17	23	16	12	12	14	12	15	16	20	18	15	Testp				
Nitrogen Gas	NM3	18	23	16	14	15	16	17	16	17	19	12	16	test6				

❖ Other Losses

Startup Losses																Other Losses	Shutdown Rate (Per Hr)	Receipte	
																Save			
Particu...	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark					
▼ Raw Material																			
PROPYLENE- C...	MT		11.2	12.3	11.5	12	12	12	12	11	11	13	12	12	test1				
ETHYLENE	MT		14.3	14.3	14.3	14.3	14.3	14.3	14.3	14.3	14.3	14.3	14.3	14.3	Test2				
PROPYLENE (P...	MT		15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	Test3				
▼ Utility Consumption																			
Power_Dis	KWH		16.7	16.7	16.7	16.7	16.7	16.7	16.7	16.7	16.7	16.7	16.7	16.7	Test4				
Water	M3		15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	Test5				
COMPRESSED ...	NM3		15.3	12.3	13.6	14.2	18.7	18.7	12.9	13.4	15.9	12.8	11.8	15.9	Test6				
Nitrogen Gas	NM3		15.3	12.3	13.6	14.2	18.7	18.7	12.9	13.4	15.9	12.8	11.8	15.9	Test7				
Effluent Treated	M3		15.3	12.3	13.6	14.2	18.7	18.7	12.9	13.4	15.9	12.8	11.8	15.9	Test8				
Cooling Water 2	KM3		15.3	12.3	13.6	14.2	18.7	18.7	12.9	13.4	15.9	12.8	11.8	15.9	Test9				
D M Water	M3		15.3	12.3	13.6	14.2	18.7	18.7	12.9	13.4	15.9	12.8	11.8	15.9	Test6				
LP Steam_Dis	MT		15.3	12.3	13.6	14.2	18.7	18.7	12.9	13.4	15.9	12.8	11.8	15.9	Test6				

❖ Shutdown rate (Per Hr)

Startup Losses	Other Losses	Shutdown Rate (Per Hr)	Receipe
Save			
Particulars	Values	Remark	
Raw Material			
PROPYLENE- CHEMICAL GRADEm		12 test	
PROPYLENE (PURITY 90% & ABOVE		21 test	
ETHYLENE		11 test	
Utility Consumption			
Cooling Water 2		12 test	
Nitrogen Gas		12 test	
COMPRESSED AIR		11 test	
Effluent Treated		11 test	
LP Steam_Dis		11 test	
Power_Dis		33 test	
Water		11 test	
D M Water		22 test	

❖ Recipe

Startup Losses	Other Losses	Shutdown Rate (Per Hr)	Receipe				
Save							
Receipe	SM100N	SRM250NC	SM60N	SR20NS	SRX100	SRM100NC	SR20NC
CAUSTIC SODA LYE – GRADE 1	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003
CHEM IRGAN-1330	0	0	0.404	1.061	0	0	0
CHEM ANTIOXIDANT 168							
CHEM CALCIUM STEARATE	0	0.53	0.25	0.53	0.53	0.53	0.53
CHEM FINASTAT 9500							
CHEM MILLAD NX8000	0	0	0	0	0	0	0
CHEM PHOSPHATE BASED CLA...	0	2.05	0.25	0	0	2.05	2.05
HYDROTALCITE DHT 4A	0	0	0.202	0.266	0	0	0
IRGANOX 1010 / ANOX 20	0	0.424	0.404	0	0.424	0.424	0.424
TINOPAL OB	0	0.00636	0	0	0	0.00636	0

2.1.1.2 Production Volume Data

The Max Capacity (MC) of NMD PP plants are derived from production captured in MIIS Portal. The list of grades will be determined based on the MIIS data for the AOP basis period. MCU data will be divided into two seasonal datasets summer and winter production, with each grade having two corresponding MCU values (one for each season).

Data to consider for each session is as follows.

- summer -1st February to 30th November
- winter -1st December to 31st January.

The MCU for each grade will be calculated by grouping production data into consecutive days—preferably 5 days, or if not available, 4, 3, 2, or 1 day. For each valid group of consecutive days, the average MCU will be calculated and then it is further averaged across all groups final average value will be considered for the next year's MCU value for that grade.

Note:

1. Production records with zero production for a grade will be excluded from the calculation.
2. Grade change days will be excluded from the consecutive day grouping.
3. The same logic will be applied separately to the winter dataset to calculate the winter MCU.

4. If any grade is not produced in winter MCU for that grade is considered from Summer data set.

With this logic All grade MC is derived for NMD PP plant for each session.

NOTE- AOP historical period basis added in configuration screen, where user can change Start date and End date to change the AOP design basis, accordingly MCU and Norms historical data will be considered for calculation.

The following logic has been formulated to derive MC values.

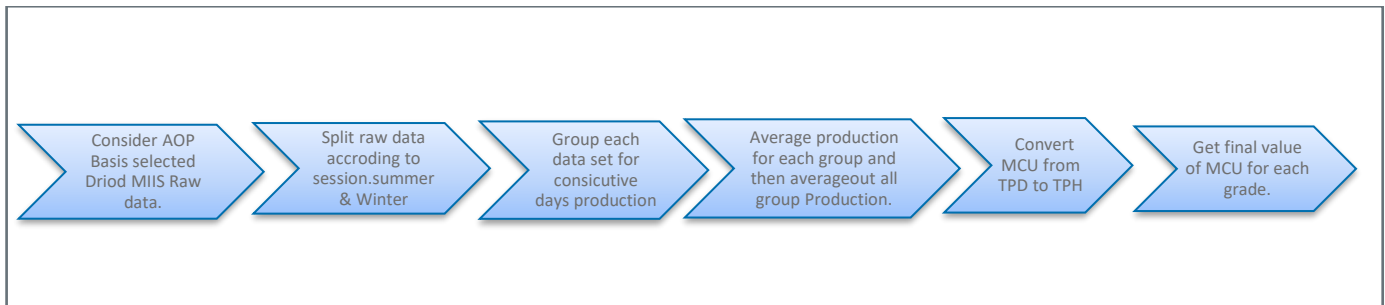


Figure 2 Digitized AOP Steps to derive MC values for NMD PP

If the user needs to modify the production rate this can be done by providing a specific reason.

The AOP digitization portal captures a month-wise percentage summary but in case of NMD PP production rate value will be same for all months for each grade and each session.

This information is captured in the AOP digitization portal as seen in the screenshot below.

Production Target													
1 2 3 4 5 6 7 8 9 10 11 12 Production & Norms Basis Shutdown/TA Activities Slowdown Activities Production Target Business Demand Maintenance Details Month wise Production plan Steady State Consumption (Norm/Quantity) Shutdown Consumption (Quantity) Slowdown Consumption (Norm/Quantity) Overall AOP Consumption (Norm/Quantity) Feed Stock Availability													
PP / NMD / PP													
Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
SM100N	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021 Click to add remark
SR20NC	14.792	14.792	14.792	14.792	14.792	14.792	14.792	14.792	14.792	13.833	13.833	14.792	14.792 Click to add remark
SR20NS	13.347	13.347	13.347	13.347	13.347	13.347	13.347	13.347	13.347	13.5	13.5	13.347	13.347 Click to add remark
SRM100NC	14.579	14.579	14.579	14.579	14.579	14.579	14.579	14.579	14.579	14.737	14.737	14.579	14.579 Click to add remark
SRM250NC	12.996	12.996	12.996	12.996	12.996	12.996	12.996	12.996	12.996	13.125	13.125	12.996	12.996 Click to add remark
SRX100	14.658	14.658	14.658	14.658	14.658	14.658	14.658	14.658	14.658	14.75	14.75	14.658	14.658 Click to add remark
SM60N	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542 Click to add remark
Percentage Summary													
Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
SM100N	100	100	100	100	100	100	100	100	100	100	100	100	100
SR20NC	100	100	100	100	100	100	100	100	100	93.52	93.52	100	100
SR20NS	98.87	98.87	98.87	98.87	98.87	98.87	98.87	98.87	98.87	100	100	98.87	98.87
SRM100NC	98.93	98.93	98.93	98.93	98.93	98.93	98.93	98.93	98.93	100	100	98.93	98.93
SRM250NC	99.02	99.02	99.02	99.02	99.02	99.02	99.02	99.02	99.02	100	100	99.02	99.02
SRX100	99.38	99.38	99.38	99.38	99.38	99.38	99.38	99.38	99.38	100	100	99.38	99.38
SM60N	100	100	100	100	100	100	100	100	100	100	100	100	100

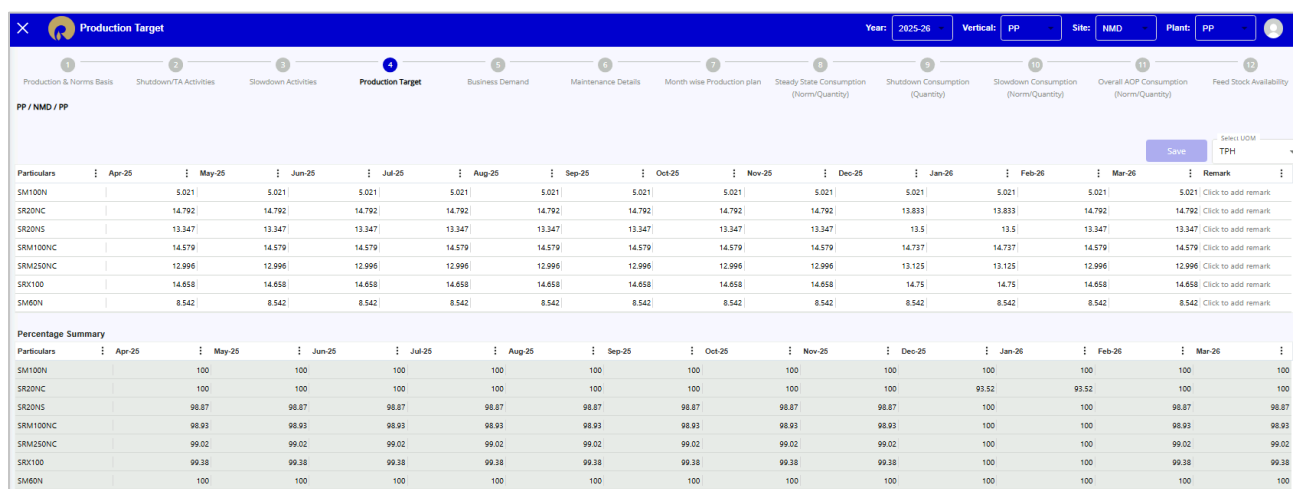
Figure 3 Digitized AOP screen for Production Volume data

The '**Calculate**' button on this screen allows the user to compute the MC value if any manual changes have been made to the AOP basis selection on the configuration screen

2.1.1.3 Business Demand

The product mix demand will be provided by the Business team and added in AOP portal manually. In the AOP digitization application, Percent monthly demand for each individual grade will be recorded.

An example of the same is mentioned in the table below.



Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
SM100N	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021	5.021	Click to add remark
SR20NC	14.792	14.792	14.792	14.792	14.792	14.792	14.792	14.792	13.833	13.833	14.792	14.792	Click to add remark
SR20NS	13.347	13.347	13.347	13.347	13.347	13.347	13.347	13.347	13.5	13.5	13.347	13.347	Click to add remark
SRM100NC	14.579	14.579	14.579	14.579	14.579	14.579	14.579	14.579	14.737	14.737	14.579	14.579	Click to add remark
SRM250NC	12.996	12.996	12.996	12.996	12.996	12.996	12.996	12.996	13.125	13.125	12.996	12.996	Click to add remark
SRX100	14.658	14.658	14.658	14.658	14.658	14.658	14.658	14.658	14.75	14.75	14.658	14.658	Click to add remark
SM60N	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542	8.542	Click to add remark

Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
SM100N	100	100	100	100	100	100	100	100	100	100	100	100	100
SR20NC	100	100	100	100	100	100	100	100	93.52	93.52	100	100	100
SR20NS	98.87	98.87	98.87	98.87	98.87	98.87	98.87	98.87	98.87	100	100	98.87	98.87
SRM100NC	98.93	98.93	98.93	98.93	98.93	98.93	98.93	98.93	98.93	100	100	98.93	98.93
SRM250NC	99.02	99.02	99.02	99.02	99.02	99.02	99.02	99.02	99.02	100	100	99.02	99.02
SRX100	99.38	99.38	99.38	99.38	99.38	99.38	99.38	99.38	99.38	100	100	99.38	99.38
SM60N	100	100	100	100	100	100	100	100	100	100	100	100	100

Figure 4 Grade-wise Business demand

Note: Details on business demand for PP to be provided by RIL.

2.1.1.4 Maintenance/Shutdown

The operations team will provide the shutdown and slowdown plan to calculate month-wise operating hours available for production.

The screenshot below depicts the shutdown activities planned, as will be captured in the AOP digitized platform.

The user must enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration (hrs) to calculate the SD-To date/time.

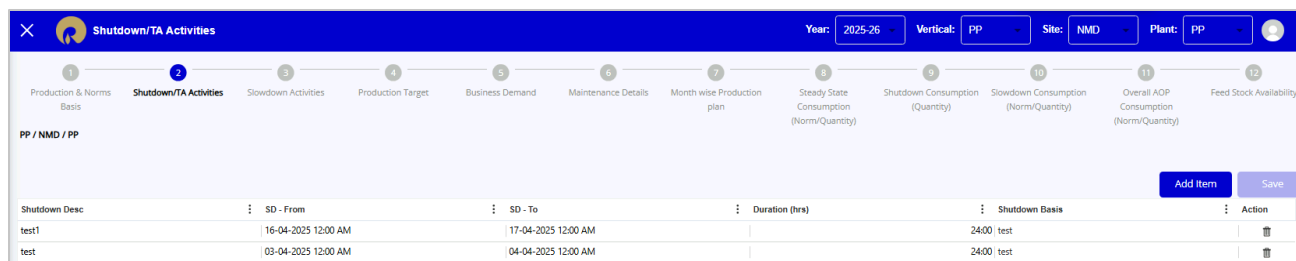


Figure 5 Digitized AOP screen for Shutdown activities

The slowdown activities can be calculated based on the slowdown rate and hours. This is called Basis Shutdown, equivalent of Slowdown.

$$\text{Basis shutdown equivalent of Slowdown hrs} = (\text{Slowdown rate} * \text{Slowdown hrs}) / \text{MCU}$$

The screen below depicts slowdown activities planned as will be captured in the digitized platform.

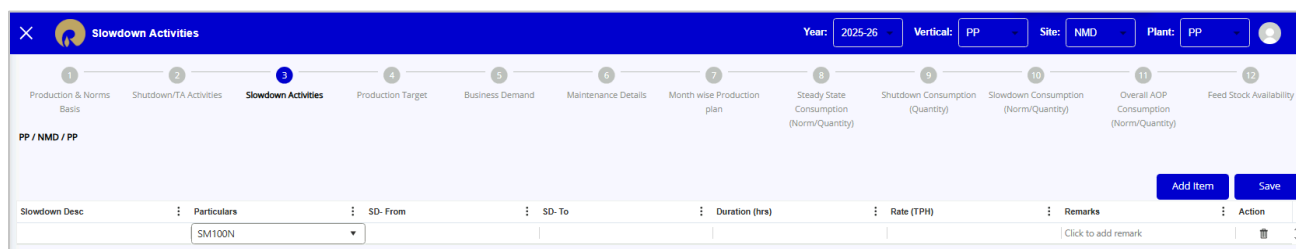


Figure 6 Digitized AOP screen for Slowdown Activities

Using this number, the operating hours available in a month will be calculated as -

$$\text{Number of Operating Hours in a month} = \text{Total Number of hrs in a month} - (\text{Shutdown Hrs} + \text{Slowdown converted to equivalent SD Hrs})$$

The screenshot below captures an example of the number of operating hours, Shutdown hours, and effective slowdown loads available in a month wise.

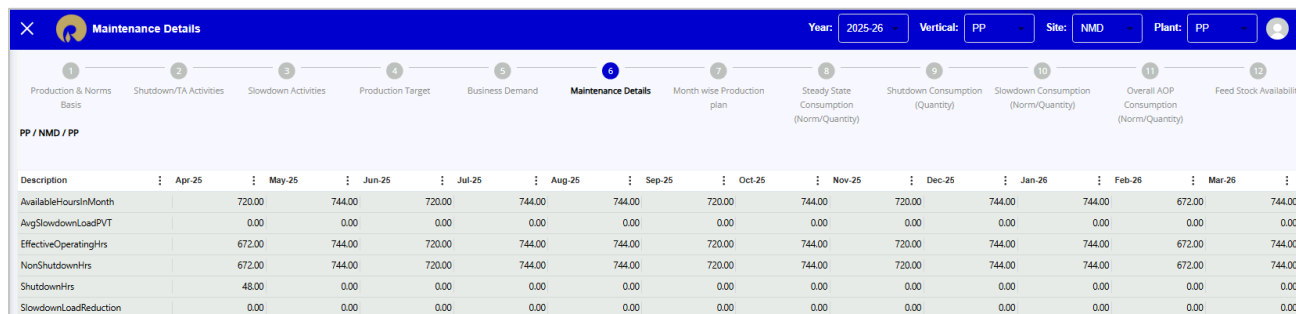


Figure 7 Number of operating hours

Note: Detail on Shutdown and slowdown activity for PP to be provided by RIL.

2.1.1.5 Calendar Year Based Production Calculations

Using the Production Volume data, business demand, and maintenance plan, the month-wise and grade-wise budgeted production quantity will be derived.

The calculation logic is as below.

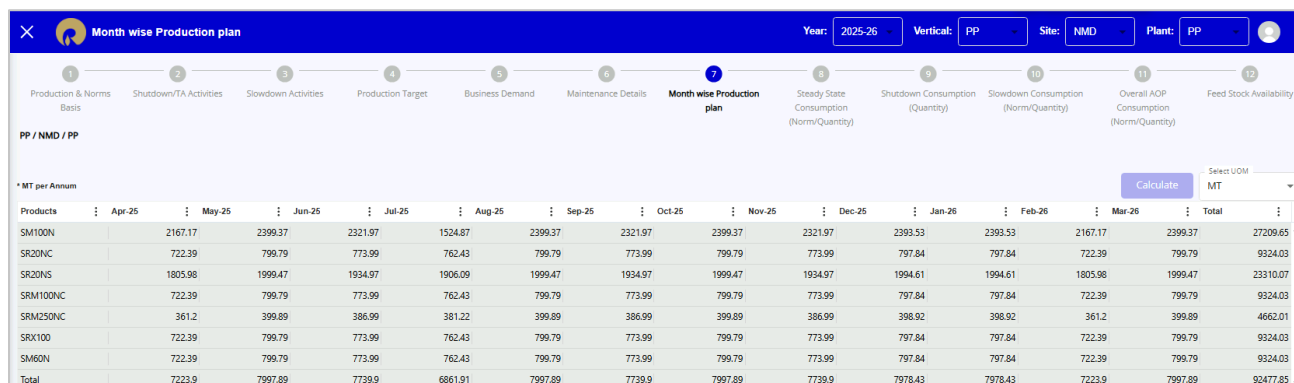
For a single grade

$$\begin{aligned} \text{Total Budgeted Monthly Production} \\ &= \text{Average MCU of All grades captured in section 2.1.1.2 (TPH)} \\ &\quad * \text{Operating hours in a month (hr)} \end{aligned}$$

$$\begin{aligned} \text{Each Grade Production for a month} &= (\text{Total Budgeted Monthly Production} * \\ &\quad \text{Percentage of Grade production demand (\%)}) / 100 \end{aligned}$$

Calendar based production calculation explained above will be used to calculate monthly production of PP.

The screenshot below captures the calculated budgeted production values.



Products	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
SM100N	2167.17	2399.37	2321.97	1524.87	2399.37	2321.97	2399.37	2321.97	2399.37	2399.37	2167.17	2399.37	27209.65
SR20NC	722.39	799.79	773.99	762.43	799.79	773.99	799.79	773.99	797.84	797.84	722.39	799.79	9324.03
SR20NS	1805.98	1999.47	1934.97	1906.09	1999.47	1934.97	1999.47	1934.97	1994.61	1994.61	1805.98	1999.47	23310.07
SRM100NC	722.39	799.79	773.99	762.43	799.79	773.99	799.79	773.99	797.84	797.84	722.39	799.79	9324.03
SRM250NC	361.2	399.89	386.99	381.22	399.89	386.99	399.89	386.99	398.92	398.92	361.2	399.89	4662.01
SRX100	722.39	799.79	773.99	762.43	799.79	773.99	799.79	773.99	797.84	797.84	722.39	799.79	9324.03
SM60N	722.39	799.79	773.99	762.43	799.79	773.99	799.79	773.99	797.84	797.84	722.39	799.79	9324.03
Total	7223.9	7997.89	7739.9	6861.91	7997.89	7739.9	7997.89	7739.9	7978.43	7978.43	7223.9	7997.89	92477.85

Figure 8 grade wise Budget Production

Note: Above Statergy to be checked and confirmed by RIL.

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 Steady-state Operations

At a high level, the norms are categorized as below –

1. Best achieved –

Raw material, byproduct & utilities norms are generally covered under this calculation basis. Based on the Selected basis AOP Period corresponding to the MCU derived in section 2.1.1.2. High and Low range of MCU (+/- 5% of MCU) is calculated. Based on derived high and low

range of production MIIS data for selected period is filtered for consecutive days production. Consecutive days' selection is preferably 5 days, or if not available, 4, 3, 2, or 1 day. the days in which production lies within the high and low range and have consecutive days production are selected for norms calculation. Norms are calculated for each material and each day from the filtered data. respective material norms are then averaged out for Final norm value.

Below is the formulated logic for deriving the Best Achieved Norms

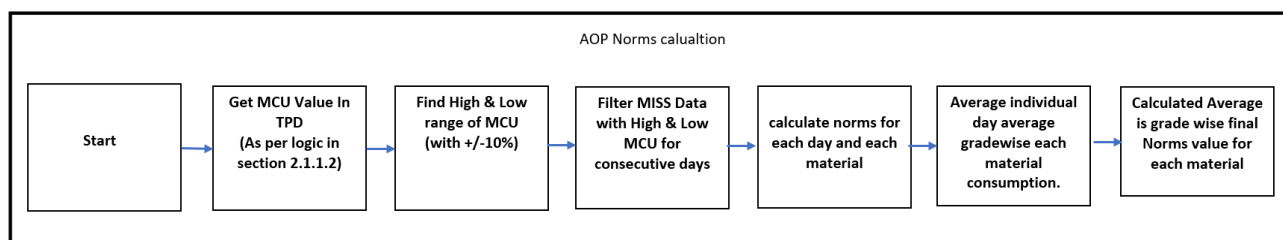
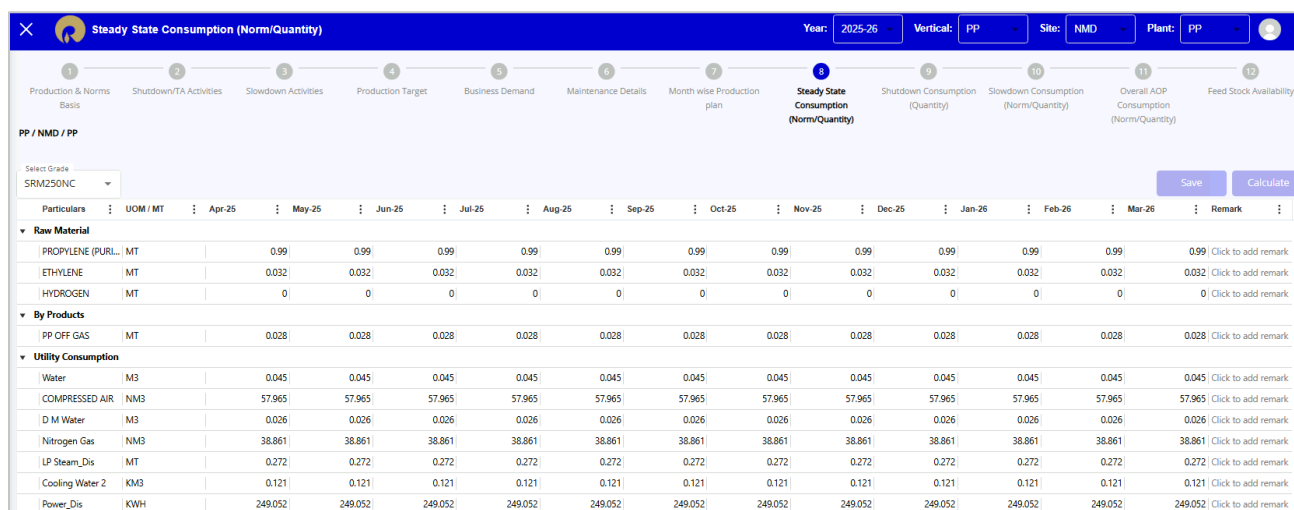


Figure 9 Norm Calculation Flow-Best Achieved

The below screenshots depict the best-achieved feed state-wise value captured in the AOP digitized platform



Particulars	UOM / MT	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
PROPYLENE (PURL...)	MT	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99 Click to add remark
ETHYLENE	MT	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032 Click to add remark
HYDROGEN	MT	0	0	0	0	0	0	0	0	0	0	0	0	0 Click to add remark
By Products														
PP OFF GAS	MT	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028 Click to add remark
Utility Consumption														
Water	M3	0.045	0.045	0.045	0.045	0.045	0.045	0.045	0.045	0.045	0.045	0.045	0.045	0.045 Click to add remark
COMPRESSED AIR	NM3	57.965	57.965	57.965	57.965	57.965	57.965	57.965	57.965	57.965	57.965	57.965	57.965	57.965 Click to add remark
D M Water	M3	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026 Click to add remark
Nitrogen Gas	NM3	38.861	38.861	38.861	38.861	38.861	38.861	38.861	38.861	38.861	38.861	38.861	38.861	38.861 Click to add remark
LP Steam_Dis	MT	0.272	0.272	0.272	0.272	0.272	0.272	0.272	0.272	0.272	0.272	0.272	0.272	0.272 Click to add remark
Cooling Water 2	KMG	0.121	0.121	0.121	0.121	0.121	0.121	0.121	0.121	0.121	0.121	0.121	0.121	0.121 Click to add remark
Power_Dis	KWH	249.052	249.052	249.052	249.052	249.052	249.052	249.052	249.052	249.052	249.052	249.052	249.052	249.052 Click to add remark

Figure 10 Digitized AOP screen for Best Achieved Norms

2. SAP Process order basis –

Catalysts and chemicals that do not have a defined formula or recipe in the DCT are typically included in this calculation method. Based on the selected AOP period, SAP data is filtered for each process order to identify the relevant catalysts and chemicals for each product grade. The consumption for each process order is then aggregated. Similarly, the production quantity for each process order is also aggregated to determine the total output. Finally, norms are calculated for each process order

based on this data.

Note: confirmation of the above understanding and final value strategy to be confirmed by RIL.

3. Recipe based Norms–

For the PP vertical, particulars catalyst chemical Norms calculation is formulated based on the Recipe provided by the user in the DCT.

The recipe is manual input to the user. The user has the provision to enter/edit the recipe value manually in the Configuration screen.

In the Normal Operations Norms screen, the cells for particulars with recipe will be read only. If the values for such particulars are to be modified, then the respective inputs should be modified in the Configuration Screen.

Example of material which have recipe-based norms are as follows

- CHEM FINASTAT 9500
- HYDROTALCITE DHT 4A
- CHEM CALCIUM STEARATE

Note: calculation for each material will be confirmed by RIL.

2.1.2.2 Shutdown Consumption.

Shutdown consumption is calculated by shutdown hours which is calculated based on shutdown schedule and shutdown consumption per hour which either constants or manual entry



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
HYDROGEN	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
PROPYLENE-CHEM...	MT	12	0	0	0	0	0	0	0	0	0	0	0	0 Test5
PROPYLENE-PURITY...	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
ETHYLENE	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
By Products														
PP OFF GAS	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
Cat Chem														
ATMER 163-ORGANIC ...	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM MILLAD N980...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
VASELINE OIL (INDU...	L	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
TINDOPAL OB	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
IRGANOX 1010 / AN...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM CALCIUM STE...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM CAT ZN 127M...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
HYDROTALCITE DHT ...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM C-ODNAR	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM CAT ZEIGLER...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CAUSTIC SODA LYE ...	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM IRGAN-1330	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM TEAL C-430	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM PROCESS OIL ...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM ANTIORIDAN...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5

Figure 11 Digitized AOP screen for Shutdown Consumption

*Shutdown Consumption = Number of shutdown hours * Shutdown consumption rate*

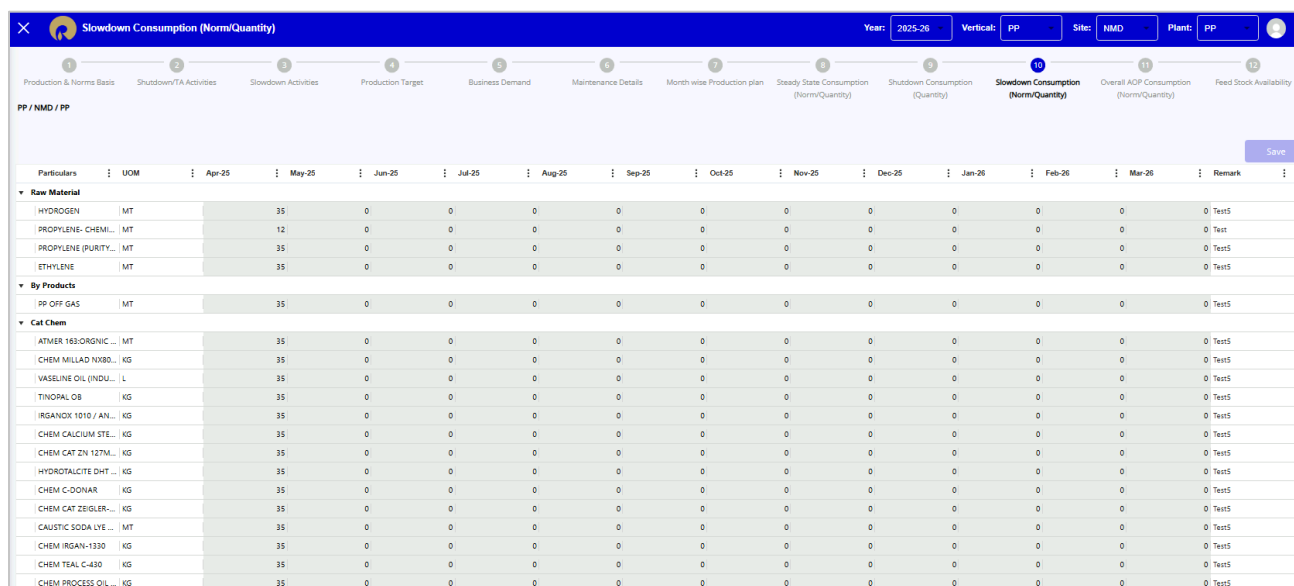
Note: Shutdown norms calculation will be confirmed by RIL.

2.1.2.3 Slowdown Consumption.

Slowdown norms are recorded for the months in which slowdown activities are planned, as specified in the Slowdown Activities screen.

The values for slowdown norms are same as the Normal Operations norms for the month.

Note- Slowdown wise norms calculation will be provided by RIL.



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
HYDROGEN	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
PROPYLENE- CHEM...	MT	12	0	0	0	0	0	0	0	0	0	0	0	0 Test5
PROPYLENE (PURITY...	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
ETHYLENE	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
By Products														
PP OFF GAS	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
Cat Chem														
ATMER 163ORGANIC ...	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM MILLAD N380...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
VASELINE OIL (INDU...	L	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
TINORAL OB	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
IRGANOX 1010 / AN...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM CALCIUM STE...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM CAT ZN 12TM...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
HYDROTALCITE DHT...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM C-ODINAR	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM CAT ZEIGLER...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CAUSTIC SODA LYE ...	MT	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM IRGAN-1330	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM TEAL C-430	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5
CHEM PROCESS OIL ...	KG	35	0	0	0	0	0	0	0	0	0	0	0	0 Test5

Figure 12 Digitized AOP screen for Slowdown Consumption

2.1.2.4 Budget Norms/ Consumption AOP

The shutdown consumption quantities will be summed with the steady-state operation norms, to derive the budgeted norms which will be inclusive of shutdown conditions. The intent here is to derive budgeted norms considering steady-state, shutdown & slowdown activities, using the logic below–

PP

Consumption Norm

$$\begin{aligned}
 &= \text{Steady State Norms as captured in section 2.1.2.1} \\
 &+ ((\text{Number of hours plant shutdown planned} * \text{Shutdown consumption per hour}) \\
 &+ \text{Startuploss} + \text{Other losses}) / \text{Total Production}
 \end{aligned}$$

3. AOP Reports

. **Note-** available reports to be sent by RIL.

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by NMD Site team